



Study Smart – Order Your VU Books
Today

0344-2023473



VU BOOK SHOP

~ Your Trusted Academic Partner ~

اپنی ورچوئل یونیورسٹی کی کتابیں آسانی، سہولت اور اعتماد کے ساتھ VU BOOK SHOP سے آرڈر کریں۔

VU BOOK SHOP پر ورچوئل یونیورسٹی کی تمام کتابیں دستیاب ہیں۔ ہم پورے پاکستان میں کیش آن ڈیلیوری، اعلیٰ کوالٹی پرنٹنگ اور بروقت ڈیلیوری کی سہولت فراہم کرتے ہیں۔ گزشتہ 6 سال سے ہزاروں مطمئن کسٹمر کا اعتماد ہماری پہچان ہے۔ مسئلہ ہونے کی صورت میں فوری تبدیلی یا ریفرنڈ



All Semester Books



Fast Delivery



100% Authentic Material

Connect & Follow Us

Connect With Us

Whatsapp Group [VU Papers & Notes Zone 2026](#)

Facebook facebook.com/vubookshop

WhatsApp wa.me/923442023473

Join our community for latest updates, offers & resources

Contact / WhatsApp

03442023473

Bio5105.

Final term

Transcription and RNA Processing

1. **What is the critical shift from a "closed" to an "open" promoter complex called?**
a) Termination b) Elongation **c) Transition to the Open Complex** d) Splicing
2. **In an open complex, the DNA is:** a) Double-stranded **b) Unwound** c) Degraded d) Replicated
3. **How many base pairs of DNA does RNA Polymerase melt during the open complex transition?** a) 5 b) 10 **c) 14** d) 20
4. **The structure formed when DNA is unwound at the transcription start site is the:** a) **Transcription Bubble** b) Replication Fork c) Lariat d) Spliceosome
5. **In prokaryotes, the unwinding of DNA during transcription does NOT require:** a) Sigma factor **b) ATP** c) RNA Polymerase d) Ribonucleotides
6. **Which factor helps stabilize melted DNA strands in prokaryotes?** a) Rho factor b) TATA box **c) Sigma factor** d) IF1
7. **Initial transcription starts when the first two ____ are brought into the active site.**
a) Amino acids **b) Ribonucleotides (NTPs)** c) Deoxyribonucleotides d) Peptides
8. **The release of short transcripts (2-9 nucleotides) during initiation is known as:**
a) Promoter escape **b) Abortive Initiation** c) Termination d) Splicing
9. **The mechanism used by the enzyme to pull DNA into itself during initiation is called:** a) Sliding b) Hopping **c) DNA Scrunching** d) Capping
10. **Promoter escape happens once the transcript exceeds how many nucleotides?**
a) 2 b) 5 **c) 10** d) 50
11. **The model suggesting RNA Polymerase is flexible and moves in a jerky way is:**
a) Sliding model b) Rolling model **c) Inchworming** d) Hopping model
12. **The speed of elongating RNA polymerase is roughly how many nucleotides per second?** a) 1-10 b) 10-20 **c) 50-100** d) 500-1000
13. **The RNA-DNA hybrid inside the transcription bubble is approximately how long?**
a) 2-3 bp **b) 8-9 bp** c) 12-14 bp d) 20-25 bp
14. **Editing where the enzyme backtracks and cuts out a sequence with a mistake is:** a) Pyrophosphorolytic editing **b) Hydrolytic editing** c) Alternative splicing d) Capping
15. **Which enzymes help manage supercoiling during transcription?** a) Helicases b) Polymerases **c) Topoisomerases** d) Ligases
16. **The sequences that trigger the release of RNA and Polymerase are called:** a) **Terminators** b) Promoters c) Enhancers d) Exons

17. **Rho protein is a ring-shaped hexameric protein that acts as an:** a) RNA polymerase **b) ATPase** c) Initiation factor d) Spliceosome
18. **The specific RNA sequence where Rho binds is called the:** a) TATA box b) Kozak sequence **c) rut site** d) Shine-Dalgarno
19. **Rho-independent termination is also known as:** a) Extrinsic termination **b) Intrinsic termination** c) Abortive initiation d) Splicing
20. **In intrinsic termination, the stable structure formed in the RNA is a:** a) Triple helix b) Lariat **c) Hairpin Loop** d) 5' Cap
21. **Which are the weakest base pairs in biology, helping the RNA release during termination?** a) **A-U** b) G-C c) G-U d) C-G
22. **Coding regions that appear in mature mRNA are called:** a) Introns **b) Exons** c) Intergenic regions d) Leaders
23. **The non-coding regions removed during RNA processing are:** a) **Introns** b) Exons c) Promoters d) Codons
24. **Where does splicing occur in a eukaryotic cell?** a) Cytoplasm **b) Nucleus** c) Ribosome d) Mitochondria
25. **Which process allows one gene to produce different proteins?** a) Transcription b) Translation **c) Alternative Splicing** d) Capping
26. **The 5' Splice Site is also known as the:** a) **Donor site** b) Acceptor site c) Branch point d) Poly-A tail
27. **The 5' Splice Site usually begins with which nucleotides?** a) AG **b) GU** c) AU d) GC
28. **The 3' Splice Site (Acceptor site) usually ends with:** a) GU **b) AG** c) AA d) UU
29. **The Branch Point is a specific ____ located upstream of the 3' site.** a) Guanine b) Cytosine **c) Adenine** d) Uracil
30. **Splicing involves two successive chemical reactions called:** a) **Transesterifications** b) Dehydration syntheses c) Hydrolysis d) Oxidations
31. **The circular/loop structure formed by the intron during splicing is a:** a) Hairpin b) Loop-d-loop **c) Lariat** d) Cloverleaf
32. **Which complexes are composed of snRNAs and proteins?** a) Ribosomes **b) snRNPs** c) Nucleosomes d) Polysomes
33. **Which snRNP recognizes the 5' splice site?** a) **U1** b) U2 c) U4 d) U6
34. **Which snRNP is considered the catalytic heart of the spliceosome?** a) U1 b) U2 c) U5 **d) U6**
35. **The active catalytic complex in splicing is the:** a) E Complex b) A Complex c) B Complex **d) C Complex**
36. **Introns that remove themselves without proteins are called:** a) **Self-Splicing Introns** b) Spliceosomal introns c) AT-AC introns d) Coding introns
37. **Self-splicing introns prove that RNA can act as an enzyme, also called:** a) Protease b) Polymerase **c) Ribozyme** d) Kinase
38. **Who earned the Nobel Prize for the discovery of ribozymes?** a) Watson and Crick **b) Thomas Cech** c) Gregor Mendel d) Charles Darwin

Genetic Code and Translation

39. **The genetic code uses how many nucleotides to specify one amino acid?** a) One
b) Two **c) Three** d) Four
40. **What does it mean that the genetic code is "Degenerate"?** a) It is broken b) It changes over time **c) More than one codon can code for the same amino acid** d) One codon codes for multiple amino acids
41. **How many possible codons are there in the genetic code?** a) 20 b) 61 **c) 64** d) 16
42. **How many codons actually code for amino acids?** a) 64 **b) 61** c) 20 d) 3
43. **Which codon is the universal "Start" codon?** a) **AUG** b) UAA c) UAG d) UGA
44. **Which amino acid is specified by the start codon AUG?** a) Valine b) Lysine **c) Methionine** d) Serine
45. **UAA, UAG, and UGA are examples of:** a) Start codons **b) Stop codons** c) Splice sites d) Promoters
46. **Where does translation occur in the cell?** a) Nucleus **b) Cytoplasm** c) Cell wall d) Mitochondria only
47. **A group of multiple ribosomes translating a single mRNA is called a:** a) Spliceosome b) Nucleosome **c) Polyribosome** d) Ribozyme
48. **The direction of translation on the mRNA is:** a) **5' to 3'** b) 3' to 5' c) N-terminus to C-terminus d) Randomized
49. **Proteins are synthesized from the N-terminus to the:** a) 5' end b) 3' end **c) C-terminus** d) P-site
50. **Which RNA carries genetic information from the nucleus to the ribosome?** a) **mRNA** b) tRNA c) rRNA d) snRNA
51. **The leader sequence of mRNA is also called the:** a) 3' UTR **b) 5' UTR** c) Exon d) Intron
52. **Which type of mRNA can code for multiple proteins?** a) Monocistronic **b) Polycistronic** c) Eukaryotic d) Spliced
53. **In which organisms are transcription and translation "coupled"?** a) **Prokaryotes** b) Eukaryotes c) Humans d) Plants
54. **The Shine-Dalgarno sequence is found in:** a) **Prokaryotic mRNA** b) Eukaryotic mRNA c) tRNA d) Proteins
55. **What is the sequence of the Shine-Dalgarno site?** a) AUG b) TATA **c) AGGAGG** d) ACCAUGG
56. **What is the first amino acid in prokaryotic translation?** a) Methionine **b) N-formylmethionine (fMet)** c) Valine d) Alanine
57. **What is the 5' modification in eukaryotic mRNA?** a) Poly-A tail **b) 7-methylguanosine Cap** c) Shine-Dalgarno d) Intron
58. **How many Adenines are typically in a Poly-A tail?** a) 5-10 b) 20-50 **c) 100-250** d) 1000
59. **Eukaryotes use which sequence for translation initiation?** a) Shine-Dalgarno **b) Kozak sequence** c) TATA box d) rut site

60. Which ribosomal subunit "scans" the eukaryotic mRNA for the AUG? a) 30S b) **40S** c) 50S d) 60S
61. Which RNA acts as an "adapter" matching codons to amino acids? a) mRNA b) **tRNA** c) rRNA d) snRNA
62. The length of tRNA is typically between: a) 10-20 nucleotides b) **73-93 nucleotides** c) 100-200 nucleotides d) 1000+ nucleotides
63. Which loop of the tRNA base-pairs with the mRNA codon? a) D loop b) T loop c) **Anticodon loop** d) Variable loop
64. The 3' end of all tRNAs ends with which sequence? a) AUG b) **CCA** c) UAA d) GGG
65. tRNAs are transcribed in eukaryotes by which enzyme? a) RNA Pol I b) RNA Pol II c) **RNA Pol III** d) DNA Pol I
66. The secondary structure of tRNA is described as a: a) Double helix b) **Cloverleaf model** c) L-shape d) Hairpin
67. The enzyme that attaches an amino acid to tRNA is: a) RNA Polymerase b) **Aminoacyl-tRNA synthetase** c) Peptidyl transferase d) Ligase
68. The bond between an amino acid and tRNA is a high-energy: a) Peptide bond b) Hydrogen bond c) **Ester bond** d) Ionic bond
69. A peptide bond forms between which two groups? a) **Carboxyl and Amino** b) Hydroxyl and Phosphate c) Sugar and Base d) Carboxyl and Methyl
70. The formation of a peptide bond releases a molecule of: a) Oxygen b) Carbon dioxide c) **Water** d) ATP
71. Which site on the ribosome binds the incoming charged tRNA? a) **A Site** b) P Site c) E Site d) mRNA site
72. Which site holds the tRNA with the growing polypeptide chain? a) A Site b) **P Site** c) E Site d) T Site
73. The "empty" tRNA leaves the ribosome from the: a) A Site b) P Site c) **E Site** d) Exit tunnel
74. The Peptidyl Transferase center is located on the: a) Small subunit b) **Large subunit** c) mRNA d) tRNA

Stages of Translation

75. The assembly of the ribosome at the start codon is called: a) **Initiation** b) Elongation c) Termination d) Splicing
76. In prokaryotes, the 30S subunit binds to the: a) 5' cap b) **Shine-Dalgarno sequence** c) TATA box d) Stop codon
77. Which site is the only one where an initiator tRNA can enter directly? a) A-site b) **P-site** c) E-site d) Small subunit
78. Prokaryotes use how many initiation factors? a) 1 b) 2 c) **3** d) 10
79. Which factor prevents the large subunit from joining too early? a) IF1 b) IF2 c) **IF3** d) EF-Tu
80. Which factor escorts the initiator tRNA to the 30S subunit? a) IF1 b) **IF2** c) IF3 d) RF1

81. In eukaryotes, which factor binds the 5' cap? a) IF2 **b) eIF4E** c) EF-G d) RF3
82. The repetitive addition of amino acids to a chain is: a) Initiation **b) Elongation** c) Termination d) Processing
83. Which factor escorts the next charged tRNA to the A-site? a) EF-Tu **b) EF-G** c) IF2 d) RF1
84. Peptide bond formation is catalyzed by: a) Ribosomal proteins **b) rRNA (Ribozyme)** c) Initiation factors d) DNA polymerase
85. The movement of the ribosome forward by three nucleotides is: a) Initiation b) Transcription **c) Translocation** d) Splicing
86. Translocation requires energy from: a) ATP **b) GTP** c) CTP d) UTP
87. In prokaryotes, translocation is catalyzed by: a) EF-Tu **b) EF-G** c) RF1 d) IF3
88. In eukaryotes, the catalytic rRNA is the: a) 16S rRNA b) 23S rRNA **c) 28S rRNA** d) 5S rRNA
89. Termination occurs when the ribosome reaches a: a) Start codon **b) Stop codon** c) TATA box d) Promoter
90. Stop codons are recognized by: a) tRNAs **b) Release Factors (RFs)** c) Sigma factors d) Polymerases
91. Which factor recognizes the stop codon UAG? a) RF1 **b) RF2** c) IF1 d) EF-Tu
92. Release factors trigger the addition of ___ to the polypeptide chain. a) An amino acid **b) A water molecule** c) A phosphate d) A sugar
93. Which factor helps disassemble the ribosome after translation? a) IF3 b) EF-G **c) Ribosome Recycling Factor (RRF)** d) RF1

Protein Targeting and Mutations

94. The mechanism for transporting proteins to their destination is: a) Splicing **b) Protein targeting** c) Transcription d) Replication
95. The "address" for a protein's destination is its: a) Codon b) Anticodon **c) Signal Sequence** d) Promoter
96. Signal sequences are typically located at the: a) N-terminus **b) C-terminus** c) Middle of the protein d) 3' end
97. Proteins for the Endoplasmic Reticulum (ER) are recognized by: a) Importin **b) Signal Recognition Particle (SRP)** c) Spliceosome d) Sigma factor
98. Proteins destined for the nucleus contain a: a) Nuclear Localization Signal (NLS) **b) Stop codon** c) TATA box d) Poly-A tail
99. Which proteins help NLS-containing proteins enter the nucleus? a) Exportins **b) Importins** c) SRP d) IFs
100. A permanent change in the DNA sequence is a: a) Transcription b) Translation **c) Mutation** d) Splicing
101. A change in a single nucleotide base pair is a: a) Frameshift mutation **b) Point Mutation** c) Deletion d) Inversion
102. A mutation that results in the same amino acid is: a) Silent **b) Missense** c) Nonsense d) Frameshift

103. **A mutation that changes a codon into a Stop Codon is:** a) Silent b) Missense **c) Nonsense** d) Frameshift
104. **Which mutation shifts the entire reading frame?** a) Silent b) Missense c) Point **d) Frameshift**

Transcription Regulation

105. **A cluster of genes under one promoter in prokaryotes is an:** a) Intron b) Exon **c) Operon** d) Enhancer
106. **Proteins that block transcription are:** a) Activators **b) Repressors** c) Factors d) Enzymes
107. **Genes turned "on" by a specific molecule are part of an:** a) **Inducible system** b) Repressible system c) Constitutive system d) Operon
108. **In negative control, a repressor binds to the:** a) Promoter **b) Operator** c) Enhancer d) Exon
109. **Eukaryotic transcription involves how many different RNA polymerases?** a) 1 b) 2 **c) 3** d) 5
110. **Which RNA Pol transcribes protein-coding genes (mRNA)?** a) RNA Pol I **b) RNA Pol II** c) RNA Pol III d) RNA Pol IV
111. **A common eukaryotic promoter element located at -25 to -30 is the:** a) Shine-Dalgarno **b) TATA Box** c) rut site d) Poly-A

Chromosomes and Aberrations

112. **Large-scale changes in chromosome structure or number are:** a) Point mutations **b) Chromosomal Aberrations** c) Splicing d) Translation
113. **The process of pairing and ordering all chromosomes is:** a) Splicing **b) Karyotyping** c) Mapping d) Sequencing
114. **In karyotyping, cells are usually arrested in:** a) Prophase **b) Metaphase** c) Anaphase d) Telophase
115. **The short arm of a chromosome is called the:** a) **p arm** b) q arm c) Centromere d) Band
116. **The long arm of a chromosome is called the:** a) p arm **b) q arm** c) Centromere d) Band
117. **The condition of having an abnormal number of chromosomes is:** a) **Aneuploidy** b) Polyploidy c) Diploidy d) Haploidy
118. **Aneuploidy is usually caused by the failure of chromosomes to separate, called:** a) Replication b) Splicing **c) Non-disjunction** d) Crossing over
119. **The presence of an extra chromosome (2n+1) is:** a) **Trisomy** b) Monosomy c) Triploidy d) Deletion
120. **Down Syndrome is an example of:** a) **Trisomy 21** b) Monosomy X c) Trisomy 18 d) Polyploidy

121. **The loss of a single chromosome ($2n-1$) is:** a) Trisomy **b) Monosomy** c) Polyploidy d) Inversion
122. **Turner Syndrome is a monosomy characterized by which karyotype?** a) XXY b) XYY **c) 45, X** d) 47, XXX
123. **Having more than two complete sets of chromosomes (e.g., $3n$, $4n$) is:** a) Aneuploidy **b) Polyploidy** c) Monosomy d) Trisomy
124. **Polyploidy is very common in:** a) Humans **b) Plants** c) Insects d) Mammals
125. **Having sets of chromosomes from different species is:** a) Autopolloidy **b) Allopolloidy** c) Monosomy d) Inversion
126. **A segment of a chromosome flipped 180 degrees is an:** a) Deletion b) Duplication **c) Inversion** d) Translocation
127. **A segment moving to a non-homologous chromosome is a:** a) Deletion b) Duplication c) Inversion **d) Translocation**
128. **Which test is used to detect numerical abnormalities?** a) PCR **b) Karyotype test** c) Western blot d) Gel electrophoresis
129. **Which syndrome is characterized by the karyotype 47, XXY?** a) Down Syndrome **b) Klinefelter Syndrome** c) Turner Syndrome d) Triple X Syndrome
130. **Which syndrome is characterized by the karyotype 47, XYY?** a) Down Syndrome b) Klinefelter Syndrome c) Turner Syndrome **d) XYY Syndrome**
131. **What is the result of Triple X Syndrome (47, XXX)?** a) Always severe symptoms **b) Many have no symptoms, some learning difficulties** c) Infertility in males d) Short stature only
132. **X-Inactivation allows survival despite extra X chromosomes by:** a) Deleting them **b) Turning off the extra ones** c) Duplicating them d) Hiding them in the cytoplasm
133. **A deletion of the tip of chromosome 5 leads to:** a) Down Syndrome **b) Cri-du-chat Syndrome** c) Turner Syndrome d) Patau Syndrome
134. **What is the term for a chromosome with a centromere at the very end?** a) Metacentric b) Submetacentric **c) Telocentric** d) Acrocentric
135. **What is the term for a chromosome with a centromere in the middle?** a) **Metacentric** b) Telocentric c) Acrocentric d) Submetacentric
136. **Edward Syndrome is a trisomy of which chromosome?** a) 21 **b) 18** c) 13 d) 15
137. **Patau Syndrome is a trisomy of which chromosome?** a) 21 b) 18 **c) 13** d) 22
138. **The primary cause of most trisomies is:** a) UV radiation b) Diet **c) Maternal age and non-disjunction** d) Splicing errors
139. **Which enzyme is responsible for synthesizing RNA?** a) DNA Polymerase **b) RNA Polymerase** c) Ligase d) Helicase
140. **The region of DNA where RNA Polymerase binds is the:** a) **Promoter** b) Intron c) Terminator d) Exon
141. **In eukaryotic cells, where is the Poly-A tail added?** a) 5' end **b) 3' end** c) Middle d) Transcription start site

142. **What is the function of the 5' cap?** a) **Protection from degradation and ribosome binding** b) Splicing out exons c) Coding for amino acids d) Stopping transcription
143. **Alternative splicing occurs by joining different combinations of:** a) Introns **b) Exons** c) Promoters d) Stop codons
144. **Which RNA has a cloverleaf secondary structure?** a) mRNA **b) tRNA** c) rRNA d) snRNA
145. **Which ribosomal site does the initiator tRNA occupy during initiation?** a) A site **b) P site** c) E site d) Exit site
146. **What does the term "Universal" mean regarding the genetic code?** a) It works in space **b) Same codons code for same amino acids in almost all life** c) It never changes d) It is very long
147. **What is a "codon"?** a) **A sequence of 3 nucleotides on mRNA** b) A sequence of 3 nucleotides on tRNA c) A type of protein d) A part of the ribosome
148. **What is an "anticodon"?** a) A sequence on mRNA **b) A sequence on tRNA** c) A sequence on DNA d) A stop signal
149. **Which factor is needed for the termination of translation?** a) Sigma factor b) Initiation factor **c) Release factor** d) Elongation factor
150. **What is the primary role of rRNA?** a) Carrying the code b) Carrying amino acids **c) Structural and catalytic part of the ribosome** d) Splicing mRNA